Mining Actionable Suggestions from High-Rated Hotel Reviews: A Cross-Domain Transfer Learning Approach

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Abstract

To what extent can NLP models detect actionable improvement suggestions within high-star hotel reviews? We address this question by applying suggestion mining—a sentence-level binary classification task—to 70,085 sentences extracted from 8,571 four- and five-star TripAdvisor reviews of Marina Bay Sands, Singapore. We compare three model tiers of increasing complexity: a regex baseline, a TF-IDF with logistic regression classifier, and a BERT model with two-stage fine-tuning. All models are first trained or evaluated on the SemEval-2019 Task 9 dataset from software forums, then tested on 80 human-annotated hotel review sentences to measure cross-domain transfer. We further apply BERTopic clustering to characterise the thematic content of predicted suggestions. BERT with two-stage fine-tuning achieves the best suggestion-class F1 of 0.581 on the hotel domain, approximately doubling the cross-domain-only result (0.286). Domain adaptation proves essential: without in-domain fine-tuning, learned models drop by up to 0.43 suggestion-class F1. Thematic analysis shows that room amenities and facilities account for over a third of detected suggestions.

1 Introduction

Hotel reviews on platforms such as TripAdvisor are widely used by travellers and hotel operators alike. Reviews with four or five stars are generally assumed to express satisfaction, yet they frequently contain concrete suggestions for improvement embedded within otherwise positive text. A guest who rates a hotel five stars may still write, "I wish the breakfast had more vegetarian options." Standard sentiment analysis classifies such reviews as positive, missing the constructive content within them.

Suggestion mining is the task of identifying sentences that propose an improvement, give advice, or recommend an action (Negi et al., 2018). Framed as sentence-level binary classification, it distinguishes suggestions from non-suggestions. The task was formalised in the SemEval-2019 Task 9 shared task (Negi et al., 2019), which provided labelled data from software developer forums. That shared task demonstrated that BERT-based models achieved the best performance for in-domain suggestion detection but also revealed that cross-domain transfer—applying a model trained on one domain to another—remains a challenge (Park et al., 2019).

While SemEval-2019 Task 9 Subtask B evaluated cross-domain transfer to hotel reviews, no prior work combines domain-adapted annotation guidelines, a multi-tier model comparison, and thematic analysis for suggestion mining in high-rated hotel reviews. Hotel reviews differ from software forums in vocabulary, sentence structure, and the way suggestions are expressed: software forum users write explicit feature requests ("The API should support batch uploads"), while hotel guests embed suggestions within narrative descriptions ("The room was lovely but the pillows could be firmer"). This domain shift makes direct transfer from existing datasets non-trivial, and raises the question of how much domain-specific adaptation is needed.

Our research question is: **To what extent can NLP models detect actionable improvement suggestions within high-star hotel reviews?**

We address this question through four contributions:

- **Domain-adapted annotation guidelines** for identifying suggestions in hotel reviews, with inter-annotator agreement measured on a calibration sample of 100 sentences.
- **A three-tier model comparison**—regex baseline, TF-IDF with logistic regression (TF-IDF + LR) baseline, and BERT classifier—that quantifies the value of increasingly rich representations for suggestion detection.

Member	Contribution
Chase	Pipeline architecture, preprocessing, BERT fine-tuning, reproducibility
Clara	EDA, report drafting (Methods, Results, Data)
Leah	Annotation guidelines, BERTopic clustering, report (Intro, Related Work, Discussion)
Xiayi	Baselines (regex, TF-IDF+LR), error analysis, manual validation, presentation

- **Two-stage fine-tuning** of BERT: first on the SemEval-2019 training data, then on annotated hotel review sentences, with a cross-domain evaluation that measures the performance delta between in-domain and cross-domain settings.
- **BERTopic-based thematic analysis** of extracted suggestions, grouping them into hotel-relevant categories (room, food and beverage, service, facilities, location, value).

2 Related Work

Suggestion mining. Negi et al. (2018) formalised suggestion mining as the task of identifying sentences that express a suggestion, defined as a sentence that proposes an improvement, gives advice, or recommends a course of action. Earlier approaches relied on hand-crafted rules targeting modal verbs and imperative constructions (Negi and Buitelaar, 2017). The SemEval-2019 Task 9 shared task (Negi et al., 2019) established benchmark datasets and attracted 33 participating teams. The best-performing systems used BERT-based architectures, with the top system achieving 78.1% suggestion-class F1 on Subtask A (in-domain). Subtask B required cross-domain generalisation from software forums to hotel reviews, where performance dropped for many systems, indicating that domain shift is a core challenge in suggestion mining.

Domain adaptation and BERT fine-tuning. Pre-trained language models such as BERT (Devlin et al., 2019) learn general linguistic representations that can be adapted to downstream tasks via fine-tuning. However, fine-tuned models can be brittle when the target domain differs from the training domain. Park et al. (2019) showed that BERT is unstable for out-of-domain suggestion mining, with high variance across runs on cross-domain data. Prasanna and Seelan (2019) addressed this by applying tri-training, a semi-supervised bootstrapping approach, to label target-domain data without manual annotation, achieving 81.94% suggestion-class F1 on the cross-domain subtask. These findings motivate our two-stage fine-tuning strategy: first training on the larger SemEval dataset for task knowledge, then adapting to hotel-domain data.

Sentiment analysis versus suggestion mining. Suggestion mining is distinct from sentiment analysis. A five-star review is positive in sentiment but may contain sentences that propose improvements. Aspect-based sentiment analysis identifies opinions about specific features (e.g. "the pool was clean"), but does not distinguish between descriptive praise and constructive suggestions (e.g. "they should heat the pool"). Our work targets this gap by extracting the subset of sentences that contain improvement-oriented content, regardless of overall sentiment, and grouping them thematically using BERTopic (Grootendorst, 2022) to produce actionable insights for hotel management.

3 Data

3.1 Marina Bay Sands Review Dataset

We collected 10,232 English-language reviews of Marina Bay Sands, Singapore from TripAdvisor, spanning January 2014 to October 2022. We filtered for four- and five-star reviews, targeting reviews likely to contain improvement suggestions embedded within positive text. After removing null entries, 8,571 reviews remained: 2,414 four-star and 6,157 five-star.

We segmented reviews into sentences using the spaCy `en_core_web_sm` model (Honnibal et al., 2020) and applied three sequential cleaning steps to the 72,391 segmented sentences: (1) a language filter removing 1 non-English sentence, (2) exact deduplication removing 1,137 duplicates, and (3) a length filter removing 1,168 sentences outside the 4–100 token range. After all preprocessing, 70,085 sentences remained, with a median length of 15 tokens and a mean of 8.4 sentences per review. Table 1 summarises the dataset statistics.

3.2 SemEval-2019 Task 9 Dataset

We use the publicly available SemEval-2019 Task 9 Subtask A dataset (Negi et al., 2019), collected from the Uservoice software feedback forum. The SemEval training set contains 8,500 sentences

Table 1: MBS dataset statistics.

Statistic	Value
Raw reviews	10,232
After 4–5 star filter	8,571
Four-star reviews	2,414
Five-star reviews	6,157
Sentences after preprocessing	70,085
Median sentence length (tokens)	15
Mean sentences per review	8.4

Table 2: SemEval-2019 Task 9 dataset statistics.

Split	Total	Sugg.	Non-sugg.	Rate
Training	8,500	2,085	6,415	24.5%
Test	833	87	746	10.4%

(2,085 suggestions, 6,415 non-suggestions; 24.5% suggestion rate). The SemEval test set contains 833 sentences (87 suggestions, 746 non-suggestions; 10.4% suggestion rate). This class imbalance, particularly in the SemEval test set, motivates our use of suggestion-class F1 rather than accuracy as the primary evaluation metric.

The SemEval dataset differs from our MBS sentences in several respects. Software forum posts tend to be explicit feature requests using modal verbs ("The system should allow…"), while hotel review suggestions are often implicit, embedded within descriptive or comparative language ("I wish the breakfast had…"). Sentence lengths and vocabulary also differ: the SemEval data contains technical terms absent from hospitality reviews. These differences constitute the domain shift that our cross-domain evaluation measures.

3.3 Annotation

We developed annotation guidelines adapted from the SemEval-2019 task definition (Negi et al., 2019, 2018), with extensions for hotel-domain edge cases such as comparative suggestions, implicit improvement requests, and the distinction between complaints and concrete suggestions.

Calibration round. All four annotators independently labelled the same MBS calibration sample of 100 sentences. To increase the proportion of informative examples, we used signal enrichment: 50% of the sample was drawn from sentences containing suggestion-signal keywords (modal verbs, imperatives, conditionals), and 50% was drawn randomly. Inter-annotator agreement was fair: Fleiss' $\kappa = 0.379$. Pairwise Cohen's κ ranged from 0.212 to 0.672 (mean 0.426). The highest pairwise agreement (0.672) occurred between two annotators who most consistently distinguished complaints from suggestions; the lowest (0.212) reflected differing thresholds for implicit suggestions. The moderate overall agreement is consistent with the subjective nature of suggestion boundaries reported by Negi et al. (2018).

Disagreement resolution. Sentences with majority agreement (3 or 4 of 4 annotators) received the majority label. Of the 100 calibration sentences, 11 produced 2-vs-2 ties; these were resolved by group discussion. Most ties involved sentences where negative sentiment co-occurred with modal verbs (e.g. "I would definitely stay again"), which annotators initially split on. After calibration, guidelines were revised to clarify two recurring edge cases: complaints without actionable direction and informational statements versus guest-directed advice.

Split annotation. After calibration, 300 additional sentences were divided into four batches of 75, with each annotator labelling one batch independently. Combined with the 100 calibration sentences, the MBS annotated set contains 400 sentences: 73 suggestions (18.25%) and 327 non-suggestions (81.75%). The suggestion rate is lower than the SemEval training set (24.5%), reflecting the less directive language typical of hotel reviews.

Train/test split. The MBS annotated set was split 80/20 (stratified by label) into an MBS train split of 320 sentences (58 suggestions, 18.1%) and an MBS test split of 80 sentences (15 suggestions, 18.8%).

4 Methods

4.1 Task Formulation

We framed suggestion mining as binary sentence classification: each sentence was labelled as suggestion (1) or non-suggestion (0). Classification operated at the sentence level, consistent with the SemEval-2019 task definition. We did not use review-level context; each sentence was classified independently.

4.2 Tier 1: Regex Baseline

The regex baseline classified a sentence as a suggestion if it matched any of 24 hand-crafted regular expression patterns. These patterns covered three categories: modal suggestions (e.g. `\bshould\b`, `\bcould improve\b`, `\bneed to\b`), imperatives (e.g. `\bplease\b`, `\brecommend\b`,

`\bconsider\b`), and conditionals (e.g. `\bi wish\b`, `\bit would be nice\b`, `\bhopefully\b`). Matching was case-insensitive. This baseline established a performance floor and demonstrated the limitations of surface patterns—for instance, "We would recommend this hotel to anyone" matches the modal pattern but is not a suggestion.

4.3 Tier 2: TF-IDF + LR Baseline

The TF-IDF + LR baseline used a TF-IDF vectoriser (unigrams and bigrams, maximum 10,000 features, minimum document frequency 2, sublinear term-frequency scaling) followed by logistic regression (regularisation $C = 1.0$, balanced class weights, maximum 1,000 iterations), implemented in scikit-learn (Pedregosa et al., 2011). The model was trained on the SemEval training set. Balanced class weights addressed the class imbalance by upweighting the minority suggestion class. This baseline showed the value of learned features over handcrafted rules while remaining interpretable and fast to train.

4.4 Tier 3: BERT Classifier with Two-Stage Fine-Tuning

Our primary model was `bert-base-uncased` (Devlin et al., 2019), a 110-million-parameter pretrained transformer implemented via the HuggingFace Transformers library (Wolf et al., 2020), fine-tuned for binary classification with an additional linear output layer.

Stage 1: SemEval fine-tuning. We fine-tuned on the SemEval training set (8,500 sentences), using a 90/10 stratified train/validation split. Sentences were tokenised with a maximum length of 128 tokens. Training used the AdamW optimiser (Loshchilov and Hutter, 2019) with learning rate 2×10^{-5}, batch size 16, 3 epochs, linear warmup over 10% of training steps, and weight decay 0.01. We selected the checkpoint with the best macro F1 on the validation split. On the SemEval validation split, BERT stage-1 achieved macro F1 of 0.886 and suggestion-class F1 of 0.828.

Stage 2: MBS domain adaptation. Starting from the stage-1 checkpoint, we further fine-tuned on the MBS train split, using an 85/15 stratified train/validation split. The learning rate was reduced to 1×10^{-5} to avoid overwriting task knowledge learned in stage 1. All other hyperparameters remained the same. This two-stage approach followed the principle of progressive fine-tuning: pretrained representations → task-specific fine-tuning → domain-specific adaptation.

All training was performed on Apple Silicon (MPS backend) with a fixed random seed of 42. Hyperparameters follow Devlin et al. (2019).

4.5 Insight Generation: BERTopic Clustering

To transform binary predictions into thematic insights, we applied BERTopic (Grootendorst, 2022) to all sentences predicted as suggestions by BERT stage-2. Sentence embeddings were computed with `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019). Dimensionality reduction used UMAP (McInnes et al., 2018) (5 components, 15 neighbours, cosine metric, minimum distance 0.0). Clustering used HDBSCAN (Campello et al., 2013) (minimum cluster size 5). Topic representations were extracted via a class-based TF-IDF procedure. The number of topics was determined automatically.

As a complementary analysis, we applied keyword-based aspect grouping to the same set of predicted suggestions. Six hotel-relevant aspects were defined: room, food and beverage, service, facilities, location, and value. Each sentence was assigned to the aspect whose keywords appeared most frequently, or to "other" if no keywords matched. This provided a coarser but more interpretable categorisation than BERTopic.

4.6 Evaluation

Our primary metric is suggestion-class F1, computed on label 1 only. We also report precision, recall, and macro F1. All models are evaluated on both the SemEval test set (in-domain evaluation) and the MBS test split (cross-domain evaluation). The difference between in-domain and cross-domain suggestion-class F1—the performance delta—quantifies the domain gap. We do not perform a separate manual validation step on the full MBS predictions. This limitation is discussed in Section 6.3.

5 Results

5.1 Model Comparison

Table 3 reports suggestion-class F1 for each model on both datasets. On the SemEval test set, BERT stage-1 achieves the highest suggestion-class F1 (0.716), followed by TF-IDF + LR (0.596) and the regex baseline (0.386). Each tier improves over the

Table 3: Suggestion-class metrics. P = precision, R = recall. Δ = MBS F1 − SemEval F1.

	SemEval			MBS			
Model	P	R	F1	P	R	F1	Δ
Regex	.405	.368	.386	.421	.533	.471	+.09
TF-IDF+LR	.486	.770	.596	.333	.133	.190	−.41
BERT s-1	.624	.839	.716	.500	.200	.286	−.43
BERT s-2	.513	.908	.656	.563	.600	.581	−.08

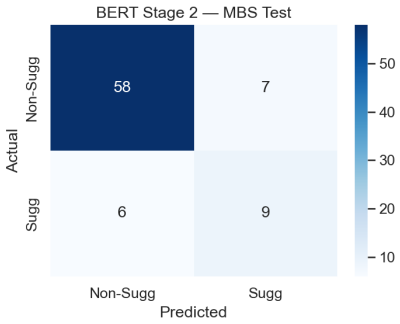

Figure 1: Confusion matrix for BERT stage-2 on the MBS test split (80 sentences).

previous, confirming that learned representations outperform hand-crafted patterns for in-domain suggestion detection.

On the MBS test split, BERT stage-2 achieves the best suggestion-class F1 (0.581), approximately doubling the stage-1 result (0.286). The two-stage fine-tuning improves both precision (0.500 → 0.563) and recall (0.200 → 0.600) on MBS, indicating that domain-specific training helps the model recognise hotel-style suggestion language. Figure 1 shows the confusion matrix for BERT stage-2 on the MBS test split.

5.2 Domain Transfer Analysis

Cross-domain performance reveals a striking pattern: the two learned models suffer large drops when moving from SemEval to MBS (TF-IDF + LR: −0.406; BERT stage-1: −0.430), while the regex baseline actually improves slightly (+0.085). This suggests that the learned models overfit to SemEval-specific vocabulary and sentence structures (e.g. software feature requests), whereas simple modal and conditional patterns transfer across domains. The differing suggestion base rates between the SemEval test set (10.4%) and MBS test split (18.8%) may also partly explain the regex improvement, as a higher base rate favours pattern-matching approaches.

However, the regex baseline's cross-domain robustness comes at the cost of low absolute performance (suggestion-class F1 of 0.471 on MBS). The regex approach generates 11 false positives on the 80-sentence MBS test split, primarily from positive modal phrases (e.g. "We would recommend this hotel to anyone") that match suggestion patterns but express praise.

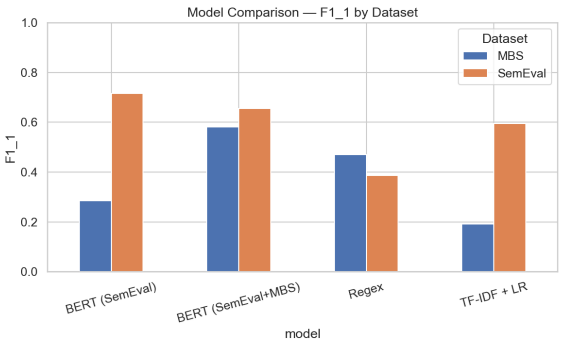

Figure 2: Suggestion-class F1 by model and dataset.

BERT stage-2 demonstrates that even a small amount of in-domain labelled data (320 sentences) can substantially recover cross-domain performance. Compared to BERT stage-1 on MBS, stage-2 reduces false negatives from 12 to 6 and improves recall from 0.200 to 0.600, at the cost of a modest increase in false positives (3 → 7). On the SemEval test set, stage-2 suggestion-class F1 drops from 0.716 to 0.656, with precision falling (0.624 → 0.513) while recall rises (0.839 → 0.908)—moderate forgetting from hotel fine-tuning. Figure 2 summarises suggestion-class F1 across models and datasets.

5.3 Suggestion Themes (BERTopic)

BERTopic analysis of 4,179 predicted suggestions from BERT stage-2 identified 56 topics, with topic 0 containing 2,278 sentences and the remaining 55 topics ranging from 5 to 56 sentences each. Topic 0 functions as a catch-all for sentences lacking distinctive sub-themes; the finer-grained topics (1–55) capture specific operational areas such as minibar pricing, elevator access, and pool amenities.

Room-related suggestions dominate (23.7%), followed by facilities (15.3%) and food and beverage (11.7%). The "other" category (31.1%) captures suggestions that do not match predefined aspect keywords. Room amenities and physical facilities together account for 39% of all detected suggestions, providing a clear priority for hotel

Table 4: Aspect distribution of predicted suggestions.

Aspect	Count	%
Other	1,299	31.1
Room	990	23.7
Facilities	638	15.3
Food & beverage	490	11.7
Value	309	7.4
Service	234	5.6
Location	219	5.2

management.

6 Error Analysis and Discussion

6.1 Error Analysis

We categorise errors from the best-performing model (BERT stage-2) on the MBS test split using keyword-based heuristics to assign each sentence to a suggestion type. Of 80 test sentences, BERT stage-2 produces 7 false positives and 6 false negatives (13 errors total).

False positives. Five of seven false positives fall in the imperative category—sentences containing words like "try", "recommend", or "please". These are borderline cases: our annotation guidelines (following the SemEval definition) classify guest-to-guest advice as suggestions, but sentences such as "We would recommend this hotel to everyone" are not improvement-oriented from the hotel's perspective. One false positive involves a positive modal phrase, and one has no keyword signal. These errors highlight the tension between the SemEval suggestion definition (any directive) and the practical goal of extracting hotel-relevant insights.

False negatives. Five of six false negatives belong to the "other" category—suggestions that lack standard keyword signals and instead express improvement needs through context-dependent phrasing. The remaining false negative is an imperative suggestion. The dominance of the "other" category among false negatives indicates that hotel review suggestions often avoid the explicit modal or imperative language that models learn to recognise from the SemEval training set.

6.2 Domain Gap Discussion

The cross-domain performance drops observed in Table 3 stem from systematic linguistic differences between software forum posts and hotel reviews. SemEval suggestions tend to be explicit feature requests ("The system should allow batch uploads"), using modal verbs and technical vocabulary. Hotel review suggestions are more often implicit, embedded within narrative praise or complaints ("I wish the breakfast had more variety"), and use hospitality-specific language absent from the SemEval training set.

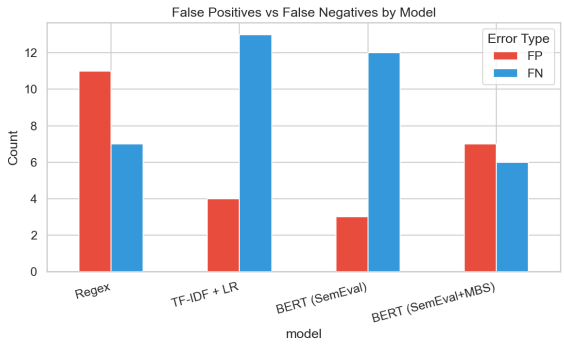

Figure 3: False positives versus false negatives by model on the MBS test split. Stage-2 fine-tuning shifts the error profile from FN-dominant to more balanced.

The learned models (TF-IDF + LR and BERT stage-1) overfit to SemEval-specific patterns, resulting in high false-negative rates on MBS (13 and 12 out of 15 suggestions missed, respectively). The regex baseline, by contrast, relies on domain-general modal and conditional patterns that transfer across domains, explaining its slight improvement on MBS (+0.085 suggestion-class F1).

BERT stage-2 partially closes the domain gap. With only 320 in-domain training sentences, it recovers suggestion-class recall from 0.200 (stage-1) to 0.600 (stage-2), while maintaining precision at 0.563. This trade-off is visible in the error comparison: stage-2 reduces false negatives from 12 to 6 but increases false positives from 3 to 7. In practice, higher recall may be preferable, as missing genuine suggestions is costlier than reviewing a few false positives. Figure 3 summarises the FP/FN trade-off across all four models.

6.3 Limitations

- **Annotation scale.** The MBS annotated set contains 400 sentences, with a test split of only 80 sentences (15 suggestions). Metrics on this test split have wide confidence intervals; a single additional correct prediction would shift suggestion-class F1 by approximately 0.05. We did not compute bootstrap confidence intervals, so reported differences should be treated as indicative rather than statistically confirmed.

- **Single hotel.** All MBS data comes from one property. Generalisability to other hotels or hospitality segments is unknown.

- **Annotation subjectivity.** Fleiss' κ of 0.379 indicates fair but not strong agreement. The boundary between complaints and suggestions is inherently fuzzy, as reflected in the 11 calibration ties. Furthermore, 300 of 400 annotated sentences were labelled by a single annotator each, with no inter-annotator validation on these split batches.

- **No human upper bound.** We did not compute human–model agreement on the test split.

- **Sentence isolation.** Each sentence is classified independently, without review-level context. Suggestions that span multiple sentences may be missed.

- **No manual validation.** We did not manually validate predictions on the full 70,085-sentence MBS dataset. The thematic analysis in Section 5.3 relies on BERT stage-2 predictions, which achieve suggestion-class F1 of 0.581 on the MBS test split but may still contain systematic errors at scale.

6.4 Future Directions

- Annotate 1,000+ sentences to reduce test-split variance.

- Include multiple hotels for cross-property generalisation.

- Investigate few-shot or prompt-based approaches using large language models as alternatives to fine-tuning.

- Use sentence-pair context to detect implicit suggestions that depend on surrounding text.

7 Conclusion

To what extent can NLP models detect actionable improvement suggestions within high-star hotel reviews? Our experiments show that suggestion mining is feasible in the hotel review domain, but cross-domain transfer from software forums degrades substantially without domain adaptation. BERT stage-2, fine-tuned on just 320 hotel-domain sentences, achieves suggestion-class F1 of 0.581 on the MBS test split—approximately double the cross-domain-only result (0.286) and the best performance among all models tested.

Our contributions are:

1. Domain-adapted annotation guidelines for hotel review suggestions, validated with Fleiss' κ of 0.379 on a 100-sentence calibration sample across four annotators.

2. A three-tier model comparison demonstrating that contextual representations (BERT) outperform bag-of-words features (TF-IDF + LR, suggestion-class F1 0.596) and surface patterns (regex, 0.386) on in-domain data, but that this advantage does not automatically transfer cross-domain.

3. Two-stage fine-tuning that improves MBS suggestion-class F1 by 0.295 points (from 0.286 to 0.581), recovering recall from 0.200 to 0.600 with only 320 in-domain training sentences.

4. Thematic analysis of 4,179 predicted suggestions (from BERT stage-2, suggestion-class F1 = 0.581) indicating that room-related suggestions account for the largest share (23.7%), followed by facilities (15.3%) and food and beverage (11.7%). At precision 0.563, roughly 44% of inputs to BERTopic may be false positives, so these distributions are approximate.

Future work should scale annotation to more hotels and review platforms, investigate prompt-based suggestion detection using large language models, and explore multi-label classification of suggestion types (explicit, implicit, comparative).

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A Annotation Guidelines Summary

Each sentence is labelled as suggestion (1) or non-suggestion (0). The core decision rule is: "Does this sentence propose a concrete action the hotel could take to improve?" Guidelines cover six edge-case categories:

1. **Wishes and desires** (label 1): "I wish the pool was open later" implies extending pool hours.

2. **Implicit comparisons** (label 1): "Other hotels offer free shuttle" implies adding a shuttle service.

3. **Conditional praise** (label 1): "Great hotel, but they need to fix the elevators" contains an actionable clause.

4. **Complaints without direction** (label 0): "The check-in took over an hour" describes a problem without proposing a fix.

5. **Positive modals** (label 0): "I would definitely stay again" uses "would" for praise, not suggestion.

6. **Guest advice** (label 1): "Try the laksa at the food court" is a recommendation with a directive element.

Guidelines were revised (v1.0 → v1.1) after the calibration round to clarify the complaint-versus-suggestion boundary and the distinction between informational statements and guest-directed advice.

B Code Availability

All code, data processing scripts, and trained model checkpoints are available at https://github.com/LanscorTK/NLP_Team6. The full pipeline can be reproduced with `python run_pipeline.py`.

C AI Tool Acknowledgement

We used Claude (Anthropic) as an assistive tool for coding support and project planning throughout this project. Specific uses included: debugging Python pipeline code, drafting boilerplate LaTeX structure, and literature search assistance via arXiv and Semantic Scholar API integrations. All substantive research decisions—task formulation, annotation guideline design, model selection, hyperparameter choices, and interpretation of results—were made by the team. Claude was not used to generate report prose; all analytical writing was authored by team members. This disclosure follows UCL guidance on acknowledging the use of generative AI tools in assessed work.